Zilong Zhang

Ph.D. Student in Statistics and Machine Learning, Georgia State University
+1 313-797-7168 zzhang52@gsu.edu Google Scholar LinkedIn GitHub Website

EDUCATION

Ph.D. in Machine Learning & Data Science **Aug. 2025 – Present**
Department of Mathematics & Statistics / Georgia State University *Atlanta, GA*

M.S. in Statistics **Aug. 2023 – May 2025**
Department of Statistics & Probability / Michigan State University *East Lansing, MI*

Bachelor of Financial Mathematics **Sep. 2019 – Jul. 2023**
School of Statistics / University of International Business and Economics *Beijing, China*

PROFILE

- Statistics Ph.D. researcher trained in financial mathematics, statistical modeling, and high-dimensional methods.
- Research interests include functional data analysis (FDA), dependence in functional time series, long-run covariance estimation, and quantitative modeling.

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Weiyi He, Junxi Zhang, Lei Ding, Chi-Kuang Yeh. "AURA: Adaptive Uncertainty-aware Refinement for LLM-as-a-Judge Auditing" *Under review, arXiv:2606.19714*, 2026.
- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. "Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning" *Under review, arXiv:2606.19057*, **2026**.

RESEARCH AND OTHER PROJECTS

Functional Data Analysis for Long-Run Dependence **Sept. 2025 – Present**

- Research on bandwidth selection for functional time series and long-run covariance estimation.

Genome-Wide Association Analysis of Treatment-Resistant Hypertension **Sept. 2024 – Dec. 2024**

- Analyzed high-dimensional genetic data from CKD and ARTH patient cohorts in an exploratory genome-wide association study of treatment-resistant hypertension.

WORK EXPERIENCE

China Securities Co., Ltd. **Jul. 2021 – Nov. 2021**
Investment Management Intern *Beijing, China*

- Supported listing work for two client companies; prepared industry surveys, risk assessments, listing materials, and internal review reports, and helped plan fundraising activities.

Chuzhou Longyun Financial Consulting Co. Ltd. **Mar. 2021 – Jun. 2021**
General Manager Assistant *Chuzhou, China*

- Provided recurring financial-audit and accounting support; assisted with corporate due diligence and client-review documentation.

Bank of Communications Chuzhou Branch **Dec. 2020 – Feb. 2021**
Account Manager Intern *Chuzhou, China*

- Supported accounting-risk monitoring, maintained credit databases, and prepared creditworthiness assessments and analysis reports.

TECHNICAL SKILLS

Programming and data: Python, R, C; NumPy, pandas, scikit-learn
Statistical methods: FDA, functional time series, covariance estimation, high-dimensional screening
Tools: Git, Linux, Jupyter, VS Code